MacEwan UNIVERSITY

CMPT 315

Lab 2 – CSS Basics, Colors, Cascading, and Multi-Page Navigation

General Information

- **Due date**: Due **@10:59 AM on 29-Sep-2026** (right before your next lab)
- **Lab weight**: 2%

OBJECTIVES:

- You will build a small multi-page travel blog about Edmonton and style it using the CSS Box Model, selectors, hover states, and CSS variables.
- You will also practice cascading and hierarchy by applying targeted styles using class and descendant selectors.

BACKGROUND:

- **The CSS Box Model:**
  - Every element is a box
  - Content: text or image
  - Padding: space inside the border
  - Border: outline around content + padding
  - Margin: space outside the border
  - You will use this to build button-like navigation, card layouts, and tag pills.
- **Selectors and Pseudo-classes:**
  - Element selector: p { ... }
  - Class selector: .card { ... }
  - Descendant selector: .site-nav a { ... }
  - Pseudo-class: .site-nav a:hover { ... }
- **Cascading and Hierarchy**
  - When styles conflict, check:
    - Inheritance (some properties inherit from parents)
    - Selector match (does it target the element?)
    - Specificity (more specific selector wins)
    - Location (later rules win if specificity is equal)
- **Colors and Variables**
  - Use :root to define variables once and reuse them everywhere using var(--name). This keeps the theme consistent and easy to change.

PART 1: Setting Up the Project

Complete these steps before you start coding.

1. Create a folder: edmonton-travel-blog
2. In VS Code: File -> Open Folder -> select edmonton-travel-blog
3. Create files: **index.html**, **city.html**, **video.html**, **contact.html** and **style.css**

PART 2: index.html (Home/blog)

*Copy/paste the following into **index.html**, then complete TODO items (HTML #1 to HTML #3).*

HTML

```
<!DOCTYPE html>
<html lang="en">

<head>
    <!-- Character encoding -->
    <meta charset="UTF-8" />
    <!-- Responsive scaling for mobile -->
    <meta name="viewport" content="width=device-width, initial-scale=1" />

    <title>Edmonton Travel Blog | Home</title>

    <!-- All pages share ONE stylesheet -->
    <link rel="stylesheet" href="style.css" />
</head>

<body>
    <!-- Container centers the site and controls max-width -->
    <div class="container">

        <header class="site-header">
            <h1>Edmonton Travel Blog</h1>
            <p>Local highlights, short itineraries, and must-see spots in Edmonton.</p>

            <!-- Navigation is a list; CSS will turn it into a nav bar -->
            <nav class="site-nav">
                <ul>
                    <li><a class="active" href="index.html">Home</a></li>
                    <li><a href="city.html">About the City</a></li>
                    <li><a href="video.html">Video</a></li>
                    <li><a href="contact.html">Contact</a></li>
                </ul>
            </nav>
        </header>

        <!-- Two-column layout: main content + sidebar -->
        <main class="layout">
            <section class="main">

                <article class="card">
                    <h2>River Valley Walk: Easy Views, Big Payoff</h2>
                    <div class="meta">Best for: sunsets • Time: 60–90 minutes</div>
                    <p>
                        Edmonton’s river valley has long trails, viewpoints, and peaceful spots
                        close to downtown. Start with a short loop and bring a warm layer—weather
                        changes quickly.
                    </p>
                    <div class="tags">
```

```
                    <span class="tag">Outdoors</span>
                    <span class="tag">Scenic</span>
                    <span class="tag">Beginner-friendly</span>
                </div>
            </article>

            <article class="card">
                <h2>Whyte Avenue Evening: Food, Shops, and People-Watching</h2>
                <div class="meta">Best for: cafes • Time: 2–3 hours</div>
                <p>
                    Whyte Avenue is a classic for browsing local shops and finding great food.
                    Pick one dessert place, one dinner spot, and walk the strip.
                </p>
                <div class="tags">
                    <span class="tag">Food</span>
                    <span class="tag">Local</span>
                    <span class="tag">Night out</span>
                </div>
            </article>

            <!-- TODO (HTML #1):
                Add ONE more blog post as <article class="card"> with:
                - a place in Edmonton (example: Muttart Conservatory, Fort Edmonton Park,
                    Royal Alberta Museum)
                - a <div class="meta"> line
                - at least TWO <span class="tag"> tags
            -->

            <!-- TODO (HTML #2):
                Add another card at the bottom:
                - heading and 2–3 sentences describing what the blog is about
            -->

        </section>

        <aside class="sidebar">
            <div class="card">
                <h3>Quick Plan (Half Day)</h3>
                <ul>
                    <li>Start: River Valley viewpoint</li>
                    <li>Lunch: Whyte Ave</li>
                    <li>End: Museum or Conservatory</li>
                </ul>
            </div>

            <div class="card">
                <h3>Tips</h3>
                <ul>
                    <li>Dress in layers</li>
                    <li>Check hours before you go</li>
                    <li>Bring a portable charger</li>
                </ul>
            </div>

            <!-- TODO (HTML #3):
                Add one more sidebar card:
                - Title: Getting Around
                - List with 2/3 items (walk, transit, rideshare, etc.)
            -->

        </aside>
```

```
        </main>

        <footer class="site-footer">
            <p>&copy; 2026 Edmonton Travel Blog (Lab Practice)</p>
        </footer>

    </div>
</body>

</html>
```

PART 3: city.html (About the City)

*Copy/paste the following into **city.html**, then complete TODO items (HTML #4 to HTML #7).*

HTML

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <title>Edmonton Travel Blog | About the City</title>
    <link rel="stylesheet" href="style.css" />
</head>

<body>
    <div class="container">
        <!-- TODO (HTML #4):
            HEADER SECTION (copy the same structure from index.html)
            1) Create a <header> with class="site-header"
            2) Add an <h1> for the site title: Edmonton Travel Blog
            3) Add a <p> that describes the current page (example: About the City)
            4) Add the navigation menu using <nav class="site-nav"> with a <ul> list
                - Each menu item is a <li> containing an <a> link
                - Use the SAME menu on every page for consistency
                - Only change which link has class="active" (the current page)
        -->

        <main class="layout">
            <section class="main">
                <div class="card">
                    <h2>Edmonton at a Glance</h2>

                    <p>
                        Edmonton is a vibrant city that combines natural scenery with urban
                        attractions, making it a great destination for short trips or weekend
                        travel. The river valley offers extensive trails and viewpoints that
                        feel surprisingly close to downtown. The city also has lively
                        neighbourhoods with local restaurants, cafés, and shops that are easy
                        to explore on foot.
                    </p>
                    <h3>What Visitors Enjoy</h3>

                    <!-- TODO (HTML #5):
                        Add a <ul> list with 4 items.
                    -->

                    <!-- TODO (HTML #6):
                        Create a header stating “Best Time to Visit” and Write 2–4 sentences.
```

```
                    -->

                    <div class="tags">
                        <span class="tag">Parks</span>
                        <span class="tag">Festivals</span>
                        <span class="tag">City life</span>
                    </div>
                </div>

                <!-- TODO (HTML #7):
                    Add a second card titled “One-Day Itinerary” containing an <ol> list with
                    4 steps.
                -->

            </section>
            <aside class="sidebar">
                <div class="card">
                    <h3>Student Practice Notes</h3>
                    <ul>
                        <li>This is a “static” informational page.</li>
                        <li>Use headings and lists for readability.</li>
                        <li>Do not copy articles from the internet.</li>
                    </ul>
                </div>
            </aside>
        </main>
        <footer class="site-footer">
            <p>&copy; 2026 Edmonton Travel Blog (Lab Practice)</p>
        </footer>
    </div>
</body>

</html>
```

PART 4: video.html (YouTube embed + attribution)

*Copy/paste the following into **video.html**, then complete TODO items (HTML #8 to HTML #9).*

HTML

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <title>Edmonton Travel Blog | Video</title>
    <link rel="stylesheet" href="style.css" />
</head>

<body>
    <div class="container">
        <header class="site-header">
            <h1>Edmonton Travel Blog</h1>
            <p>Video highlight</p>
            <nav class="site-nav">
                <ul>
                    <li><a href="index.html">Home</a></li>
                    <li><a href="city.html">About the City</a></li>
                    <li><a class="active" href="video.html">Video</a></li>
                    <li><a href="contact.html">Contact</a></li>
```

```
                </ul>
            </nav>
        </header>
        <main class="layout">
            <section class="main">
                <div class="card" id="featured-video">
                    <h2>Edmonton Travel Video</h2>
                    <p class="notice">
                        Replace <strong>VIDEO_ID_HERE</strong> with a real YouTube video ID.
                        Then provide attribution to the creator (Standard YouTube License).
                        We haven't covered &lt;iframe&gt; in class yet but this is good practice.
                    </p>
                    <div class="video-wrap">
                        <!-- TODO (HTML #8):
                            Add a YouTube video using an <iframe> embed. Take the video ID from a
                            normal YouTube link then place it into the embed format. For example,
                            if the link is https://www.youtube.com/watch?v=EfJo3MHFkFI, the video
                            ID is EfJo3MHFkFI, so your iframe src becomes
                            https://www.youtube.com/embed/EfJo3MHFkFI. Keep the iframe attributes
                            (title, allow, allowfullscreen) so the player works correctly. After
                            the iframe, include attribution by writing the creator's name and
                            adding a working link to the video or creator's channel; if the video
                            cannot be embedded (it may say playback is disabled on other websites),
                            choose a different YouTube video that allows embedding.
                        -->
                    </div>
                    <p class="meta">
                        Video under Standard YouTube License.
                        Attribution: <strong>Creator Name Here</strong> (YouTube).
                        Link: <a href="#" target="_blank" rel="noopener">Creator Channel or
                            Video URL</a>
                    </p>

                    <!-- TODO (HTML #9):
                        1)Replace Creator Name Here with the creator's name.
                        2)Replace # with the real URL (creator channel OR the video page).
                    -->

                </div>
            </section>
            <aside class="sidebar">
                <div class="card">
                    <h3>How to find a Video ID</h3>
                    <p>
                        In a YouTube URL, the ID is usually after <strong>v=</strong>.
                        Example: https://www.youtube.com/watch?v=<strong>VIDEO_ID</strong>
                    </p>
                </div>
            </aside>
        </main>
        <footer class="site-footer">
            <p>&copy; 2026 Edmonton Travel Blog (Lab Practice)</p>
        </footer>
    </div>
</body>

</html>
```

PART 5: contact.html (Contact info)

*Copy/paste the following into **contact.html**, then complete TODO item (HTML #10).*

HTML

```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <title>Edmonton Travel Blog | Contact</title>
    <link rel="stylesheet" href="style.css" />
</head>

<body>
    <div class="container">
        <header class="site-header">
            <h1>Edmonton Travel Blog</h1>
            <p>Contact</p>
            <nav class="site-nav">
                <ul>
                    <li><a href="index.html">Home</a></li>
                    <li><a href="city.html">About the City</a></li>
                    <li><a href="video.html">Video</a></li>
                    <li><a class="active" href="contact.html">Contact</a></li>
                </ul>
            </nav>
        </header>
        <main class="layout">
            <section class="main">
                <div class="card">
                    <h2>Contact Us</h2>
                    <p class="notice">
                        Do not publish your personal home address or private phone number.
                        Use fictional contact info or public/organizational contact info.
                    </p>
                    <ul>
                        <li><strong>Phone:</strong> 311 (Edmonton service line)</li>
                        <li>
                            <strong>Email:</strong>
                            <a href="mailto:info@example.com">info@example.com</a>
                        </li>
                        <li><strong>Location:</strong> Edmonton, Alberta (Canada)</li>
                    </ul>

                </div>

                <!-- TODO (HTML #10):
                    Add a second card titled “Business Hours” with a list of days/times.
                -->

            </section>

            <aside class="sidebar">
                <div class="card">
                    <h3>Future Upgrade</h3>
                    <p>
                        In a later lab, you will learn how to add a contact form using HTML
                        forms and basic validation.
                    </p>
                </div>
            </aside>
        </main>
```

```
        <footer class="site-footer">
            <p>&copy; 2026 Edmonton Travel Blog (Lab Practice)</p>
        </footer>

    </div>
</body>

</html>
```

PART 6: style.css

*Copy/paste the following into **style.css**, then complete TODO items (CSS #1 to CSS #10).*

CSS

```
/* ---------------------------
   style.css (Student Tasks Version)
   Selectors + Box Model + Variables
   ---------------------------

   HOW TO USE THIS FILE:
   - Keep the selectors as they are (e.g., .site-nav a).
   Only add properties inside { }.
   - Use CSS variables with var(--name) whenever possible.
   - Use the Box Model (padding, margin, border, border-radius) to
   create spacing and “button/card” looks.
   - After each edit: Save → Refresh the browser.

   DEBUG TIP (CSS Cascade):
   If a style does not apply, check:
   1) selector matches the HTML element
   2) you saved the file
   3) no typos (property names, missing semicolons)
   4) whether another rule overrides it (order/specificity)
*/

/* ---------------------------
   CSS Variables:
   You may keep these values or adjust them to create a different theme.
   Typical values:
   - colors in hex (#RRGGBB)
   - common neutrals: light background, dark text
   - one accent color used for highlights/hover
   --------------------------- */
:root {
    --bg: #f4f4f9;
    /* background color (light) */
    --text: #222;
    /* main text color (dark) */
    --primary: #2c3e50;
    /* header/nav background color */
    --accent: #0A6C74;
    /* highlight/hover color */
    --white: #ffffff;
    /* white */
    --border: #ddd;
    /* light border color */
}

/* Base styles (inheritance applies to many text properties)
   You can adjust:
```

```
   - font-family (Arial, Helvetica, sans-serif)
   - line-height (e.g., 1.4 to 1.8)
   - background and color using variables */
body {
    margin: 0;
    font-family: Arial, sans-serif;
    background: var(--bg);
    color: var(--text);
    line-height: 1.6;
}

/* Centers page content and prevents text from stretching too wide
   You can adjust:
   - max-width (e.g., 900px to 1200px)
   - padding (e.g., 16px to 32px) */
.container {
    max-width: 1050px;
    margin: 0 auto;
    padding: 20px;
}

/* Header “banner”
   You can adjust:
   - padding (e.g., 16px–28px)
   - border-radius (e.g., 8px–16px)
   - background using var(--primary) */
.site-header {
    background: var(--primary);
    color: var(--white);
    padding: 20px;
    border-radius: 12px;
}

/* ---------------------------
   Descendant selectors:
   The UL is styled only inside .site-nav so other lists are unaffected.
   Typical values:
   - list-style: none; removes bullets
   - text-align: center; centers the menu
   --------------------------- */
.site-nav ul {
    list-style: none;
    margin: 0;
    padding: 0;
    text-align: center;
}

/* Each LI controls positioning of each menu item.
   Typical values:
   - display: inline-block; puts items side-by-side
   - margin: small horizontal spacing like 4px–12px */
.site-nav li {
    display: inline-block;
    margin: 0 6px;
}

/* TODO (CSS #1): Nav links as buttons
   Target: .site-nav a (the clickable element)

   Typical values:
   - display: inline-block so padding/background behave like a “button”.
```

```
   You should define more properties to make links look like buttons:
   - text color: usually var(--white)
   - remove underline: text-decoration (often 'none')
   - padding: increases clickable area (common: vertical 8–12px, horizontal 12–22px)
   - border-radius: rounding (common: 4px–12px; or 999px for pill style)

   Optional additions:
   - background for non-hover state (transparent or subtle)
   - font-weight (normal/bold)
   - transition: smooth hover effect (often background-color 0.2s–0.3s)
*/
.site-nav a {
    display: inline-block;
    /* add more properties here */
}

/* TODO (CSS #2): Hover effect
   Target: .site-nav a:hover

   You should define what changes when the mouse hovers:
   - background color: commonly var(--accent)
   - optionally adjust text color (if contrast becomes poor)
   - optionally change underline/border effect instead of background (design choice)
*/
.site-nav a:hover {
    /* add properties here */
}

/* TODO (CSS #3): Active link highlight
   Target: .site-nav a.active

   This should make the current page link look “selected.”
   Common approaches (choose one):
   - subtle background (semi-transparent white)
   - underline or border-bottom
   - slightly different text color
   - stronger font-weight

   Keep it visually consistent with your hover style.
*/
.site-nav a.active {
    /* add properties here */
}

/* Two-column layout (Flexbox)
   You may keep this as-is. CSS Layouts will be taught be in a future lecture
   Typical values:
   - gap: 12px–28px */
.layout {
    margin-top: 20px;
    display: flex;
    gap: 20px;
}

.main {
    flex: 2;
}

.sidebar {
    flex: 1;
```

```
}

/* TODO (CSS #4): Sidebar card look
   Target: .sidebar

   Goal: Make the sidebar area look like a clean panel.
   Typical properties:
   - background: var(--white)
   - padding: 12px–20px
   - border: 1px solid var(--border)
   - border-radius: 8px–16px

   Optional:
   - slightly different background (very light tint)
*/
.sidebar {
    /* add properties here */
}

/* TODO (CSS #5): Card styling (Box Model)
   Target: .card

   Goal: Make every card look like a readable block.
   Typical properties:
   - background: var(--white)
   - padding: 14px–20px
   - margin-bottom: 12px–18px
   - border: 1px solid var(--border)
   - border-radius: 8px–16px

   Optional:
   - box-shadow for depth (subtle; do not overdo)
*/
.card {
    /* add properties here */
}

/* TODO (CSS #6): Meta line
   Target: .meta

   Goal: Smaller, less prominent text than body content.
   Typical properties:
   - font-size: around 0.85rem–0.95rem
   - color: a muted grey (e.g., #666 or #777)
   - margin-bottom: 6px–12px
*/
.meta {
    /* add properties here */
}

/* TODO (CSS #7): Tag “pill” styling
   Target: .tag

   Typical values:
   - display: inline-block so padding/background behave properly.

   Goal: Define more properties to make tags look like pill labels
   - padding: small (vertical 4–7px, horizontal 10–16px)
   - margin: small spacing between tags
   - border: 1px solid (often #ccc or var(--border))
   - border-radius: very large (e.g., 999px) for pill shape
```

```
   Optional:
   - background: white or very light tint
   - text color: var(--primary) or var(--text)
*/
.tag {
    display: inline-block;
    /* add more properties here */
}

/* TODO (CSS #8): Video container
   Target: .video-wrap

   Goal: The iframe should appear inside a neat frame with rounded corners.
   Typical properties:
   - border: 1px solid var(--border)
   - border-radius: 8px–16px
   - overflow: hidden (important for rounded corners)
   - background: #000 (prevents white gaps around the video)
*/
.video-wrap {
    /* add properties here */
}

.video-wrap iframe {
    width: 100%;
    height: 420px;
    border: 0;
    display: block;
}

/* TODO (CSS #9): ID selector example
   Target: #featured-video
   - styles ONLY the element with id="featured-video"

   Goal: Use this to emphasize the featured section
   - this is more specific than .card
*/
#featured-video {
    /* Examples of what you CAN define here:
     - a different border style (solid/dashed)
     - a thicker border width
     - extra padding or margin
     - a subtle background tint
     - a stronger border-left accent
  */
}

/* TODO (CSS #10): Notice styling
   Target: .notice

   Goal: Make instruction boxes stand out.
   Typical properties:
   - border: dashed (e.g., 1px dashed #bbb)
   - padding: 10px–14px
   - border-radius: 8px–16px
   Optional:
   - background: white or a very light tint
   - slightly muted text color
*/
.notice {
```

```
    /* add properties here */
}

/* Footer
   You can adjust padding and color but keep it readable. */
.site-footer {
    text-align: center;
    padding: 18px 10px;
    color: #555;
}
```

PART 7: Design Studio

*After required TODOs work, choose ONE option from each category (Navigation + Cards + Layout). Replace your existing rules in **style.css** with the chosen block.*

A. Navigation options (choose 1)

CSS - A1 Button bar (default)

```
.site-nav a{
color: var(--white);
text-decoration: none;
padding: 10px 18px;
display: inline-block;
border-radius: 6px;
}
.site-nav a:hover{ background: var(--accent); }
.site-nav a.active{ background: rgba(255,255,255,0.18); }
```

CSS - A2 Minimal underline nav

```
.site-nav a{
color: var(--white);
text-decoration: none;
padding: 8px 10px;
  display: inline-block;
  border-bottom: 2px solid transparent;
}
.site-nav a:hover{ border-bottom-color: var(--accent); }
.site-nav a.active{ border-bottom-color: var(--white); }
```

CSS - A3 Pill nav

```
.site-nav a{
  color: var(--white);
  text-decoration: none;
  padding: 10px 16px;
  display: inline-block;
  border-radius: 999px;
  background: rgba(255,255,255,0.12);
}
.site-nav a:hover{ background: var(--accent); }
.site-nav a.active{ background: rgba(255,255,255,0.24); }
```

B. Card options (choose 1)

CSS - B1 Border cards (default)

```
.card{
  background: var(--white);
  padding: 16px;
  margin-bottom: 15px;
  border: 1px solid var(--border);
  border-radius: 12px;
}
```

CSS - B2 Shadow cards

```
.card{
  background: var(--white);
  padding: 16px;
  margin-bottom: 15px;
  border-radius: 12px;
  box-shadow: 0 6px 18px rgba(0,0,0,0.08);
}
```

CSS - B3 Accent left-border cards

```
.card{
  background: var(--white);
  padding: 16px;
  margin-bottom: 15px;
  border: 1px solid var(--border);
  border-left: 6px solid var(--accent);
  border-radius: 12px;
}
```

PART 8: Theme switch with CSS variables

*Paste this into **style.css:***

CSS

```
.theme-dark{
  --bg:#0f172a;
  --text:#e5e7eb;
  --primary:#111827;
  --accent:#38bdf8;
  --white:#111827;
  --border:#334155;
}
```

*Then apply to any one of your pages by changing **<body>** to:*

HTML

```
<body class="theme-dark">
```

PART 9: Debug Challenge

*Add this at the top of your **style.css**, observe what changes, then fix it with a more specific selector:*

CSS

```
a{ color: red; }
```

PART 10: Local Testing + Deployment

Before you deploy your website on GitHub, you must first confirm it works locally. Once local testing succeeds you will go ahead and deploy. You can follow these steps:

1. **Save your files**
    - Windows: **Ctrl + S**
    - Mac: **Cmd + S**
2. **Test Option A (Highly Recommended): Use Live Server in VS Code**
    - Right-click index.html → **Open with Live Server**
    - Your browser will open with a local address like http://127.0.0.1:5500/
    - Save changes and the page updates automatically (or refresh if needed)
3. **Test Option B (Simple): Open the file directly**
    - In VS Code, right-click index.html
    - Choose **Reveal in File Explorer** (Windows) / **Reveal in Finder** (Mac)
    - Double-click index.html to open it in your browser
4. **Verification checklist**
    - Nav links are horizontal and button-like.
    - Hover changes nav background.
    - Cards have padding, border, and rounded corners.
    - Sidebar looks like a card.
    - Tags look like pills.
    - Video page shows an embedded frame after you replace the video ID.
5. **Deploy to the Internet**
    - Refer to previous labs or use your preferred steps to **go live** on GitHub Pages.

SUBMISSION

Submit **one PDF document** through mêskanâs containing your answers to the questions below:

1. Paste your **GitHub Repo URL** (e.g., https://github.com/username/edmonton-travel-blog)
2. Paste your live **GitHub Pages URL** (e.g., https://username.github.io/edmonton-travel-blog)
3. Can you explain what **inline**, **embedded** and **external** styles are? Which was used for this lab?
4. What are the **7 types of selectors** covered in lecture? Were all of them present in this lab? Make sure to check the full starter code as well, not just the TODOs you completed.
5. What changes did the **debug challenge** introduce? Explain how you solved it.

DEMO: Tuesday, September 29, 2026, at 11:00 AM

- Demonstrate your completed lab during the lab session.
- Be prepared to explain your solution and answer questions about your code.

MARKING SCHEME

Component	Mark
Lab Questions (PDF Submission)	5
Lab Demo	15
Total	**20**